Family Background and Reading Processes as Predictors of Reading Performance: Evidence From Ten PISA 2018 Education Systems

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1. Introduction

Research Context

Reading literacy is central to students' participation in school and society. It involves understanding, using, evaluating, reflecting on and engaging with texts for different purposes (OECD, 2019a). PISA 2018 placed reading at the centre of its assessment, making the dataset especially suitable for examining how family background and students' reading-related characteristics are associated with reading performance.

Family resources may shape access to books, language-rich activities and learning support. PISA measures broad socioeconomic background using the index of economic, social and cultural status (ESCS), which combines parental education, parental occupational status and home possessions. A value near zero represents the level of an average student across OECD countries (OECD, 2019b). The number of books in the home provides an additional indicator of educational and cultural resources. Cross-national research has linked a family culture of books with educational success (Evans et al., 2010).

Family background alone, however, cannot capture how students approach reading. PISA 2018 includes indices for enjoyment of reading, perceived reading competence, perceived reading difficulty and knowledge of effective reading strategies. The strategy indices cover understanding and remembering information, summarising texts and assessing the credibility of information. These variables are closely connected to the processes required in the PISA reading framework, but they should be treated as associations rather than causes.

The course reading by Fuchs and Woessmann (2004) demonstrates why a multivariate approach is important. Their PISA analysis showed that a positive bivariate relationship can change when related family and school characteristics are controlled. Following that logic, this project first reports descriptive and bivariate findings and then estimates a series of multiple linear regression models.

The regression models were specified in theoretically meaningful blocks instead of selecting variables only because they produced a high R^2. The first block represents family background and education context. The second adds reading engagement and self-concept, the third adds metacognitive strategy knowledge, and the fourth adds students' reported effort during the PISA test. This structure shows how much each group of variables contributes to explained variation in reading performance.

Research Questions

RQ1. What are the distributions of reading performance, ESCS, home-book resources and the main reading-related indices in the trimmed PISA 2018 dataset?

RQ2a. Do female and male students differ in average reading performance?

RQ2b. Does average reading performance differ across the six home-book categories?

RQ3. Is ESCS linearly correlated with reading performance?

RQ4a. How much variation in reading performance is explained by a multiple linear regression model, and how does R^2 change when background and education context, reading engagement and self-concept, metacognitive strategy knowledge, and test effort are added in blocks?

RQ4b. After family background, gender, education system, grade placement and grade repetition are controlled, which reading-related predictors remain significantly associated with reading performance?

Objectives

The main objective is to examine how family background and students' reading processes jointly predict reading performance while distinguishing statistical association from causal influence.

2. Data Description

Dataset Overview

The analysis uses the trimmed PISA 2018 student dataset supplied for the course. It contains 80,016 students from ten education systems: Canada, Estonia, Finland, Hong Kong (China), Ireland, Korea, Macao (China), Poland, B-S-J-Z (China) and Singapore. The number of observations differs across systems, so the pooled dataset is not balanced by education system.

The original international PISA student files contain questionnaire responses, performance plausible values, questionnaire indices, sampling weights and replicate weights (OECD, n.d.). The course dataset is a reduced file. It includes only the first reading plausible value and does not contain standard student sampling weights, replicate weights or school identifiers. The analysis is therefore appropriate as a course exercise but should not be treated as an official population estimate.

Variables Used

Table 1
Variables used in the project

Variable	Description	Type	Role
PV1READ	Plausible Value 1 in Reading	Continuous	Dependent variable
ESCS	Economic, social and cultural status	Continuous	Background predictor
ST013Q01TA	Number of books at home, six categories	Categorical	Background predictor
ST004D01T	Student gender: female or male	Categorical	Grouping/control variable
CNT	Education-system code	Categorical	Control variable
GRADE	Grade relative to the modal grade in the system	Continuous	Control variable
REPEAT	Whether the student has repeated a grade	Categorical	Control variable
JOYREAD	Enjoyment of reading	Continuous index	Engagement predictor
SCREADCOMP	Perceived reading competence	Continuous index	Self-concept predictor
SCREADDIFF	Perceived reading difficulty	Continuous index	Self-concept predictor
UNDREM	Knowledge of understanding/remembering strategies	Continuous index	Metacognitive predictor
METASUM	Knowledge of effective summarising strategies	Continuous index	Metacognitive predictor
METASPAM	Knowledge of credibility-assessment strategies	Continuous index	Metacognitive predictor
EFFORT1	Self-reported effort in the PISA test, 1-10	Continuous	Test-engagement predictor

Data Preparation

SPSS user-missing responses were treated as missing values. PV1READ and GRADE had no missing observations. Missingness was 2.28% for ESCS, 2.20% for home books, 2.73% for JOYREAD, 4.79% for SCREADCOMP, 4.65% for SCREADDIFF, between 6.44% and 7.01% for the three metacognitive indices, and 10.25% for EFFORT1. Gender was missing for two students and grade repetition for 2,532 students (3.16%). Descriptive and bivariate analyses used all available observations. All regression models used the same complete-case sample of 63,284 students so that their R^2 values could be compared fairly.

Gender and grade repetition were recoded as two-category factors. Home books was retained as a six-category factor because its response intervals are unequal. Education system was treated as a factor, while GRADE and the questionnaire indices were retained as numerical variables. The reference categories were female, no grade repetition, 0-10 books and Canada.

3. Methodology

Descriptive Analysis

Means, medians, standard deviations, ranges, frequencies and percentages were used to describe the variables. Histograms were used to examine PV1READ and ESCS. Group means and 95% confidence intervals were used to summarise reading performance across home-book categories.

Comparative Analysis (t-tests/ANOVA)

An independent-samples t test was used to compare female and male students. A one-way ANOVA was used to compare the six home-book groups. Before both tests, the Brown-Forsythe form of Levene's test, centred at the group median, was used to check equality of variance. Because both Levene tests were significant, Welch's t test and Welch's one-way ANOVA were reported. These versions do not require equal group variances. Holm-adjusted pairwise Welch comparisons were used after the ANOVA.

Gender null hypothesis (H_0): Female and male students have equal mean reading scores.

Gender alternative hypothesis (H_1): Female and male students have different mean reading scores.

Books null hypothesis (H_0): All six home-book groups have equal mean reading scores.

Books alternative hypothesis (H_1): At least one home-book group has a different mean reading score.

Correlation Analysis

Pearson's product-moment correlation measured the direction and strength of the linear relationship between ESCS and PV1READ. Both variables are continuous, and the scatterplot showed an approximately linear positive trend. Statistical significance was assessed at the .05 level.

Correlation null hypothesis (H_0): There is no linear correlation between ESCS and reading performance in the population ($\rho = 0$).

Correlation alternative hypothesis (H_1): There is a linear correlation between ESCS and reading performance in the population ($\rho \neq 0$).

Multiple Linear Regression

Four nested regression models were estimated on the same complete-case sample. Model 1 included ESCS, home books, gender, education system, relative grade placement and grade repetition. Model 2 added reading enjoyment, perceived reading competence and perceived reading difficulty. Model 3 added knowledge of strategies for understanding and remembering, summarising, and assessing credibility. Model 4 added reported test effort and was treated as the final model.

The overall null hypothesis was that all non-intercept coefficients in the final model were zero; the alternative was that at least one coefficient was not zero. R^2 and adjusted R^2 assessed explanatory power, while nested-model F tests assessed whether each new block improved model fit. For RQ4b, individual reading-related coefficients were tested after all background and education-context variables were included.

No interaction term was included. The research questions concerned the additive associations of family background, education context, reading processes and test effort with reading performance, and no theory-driven moderation hypothesis was specified. An additive model therefore matched the stated aims and kept the blockwise R^2 comparison interpretable.

Linearity and residual patterns were examined graphically. Multicollinearity was checked using adjusted generalised variance inflation factors, and Cook's distance was used to examine influential cases. The non-constant variance test was significant, so HC3 heteroskedasticity-robust standard errors and confidence intervals were used for the final coefficient tests. All analyses were conducted in R. Because the sample is large, effect sizes and explained variance are considered together with p values.

4. Results

Descriptive Statistics

Table 2
Descriptive statistics for continuous variables

Variable	N	M	SD	Median	Minimum	Maximum
PV1READ	80,016	525.63	99.97	530.78	89.43	878.86
ESCS	78,193	0.031	0.952	0.120	-6.760	4.037
GRADE	80,016	-0.160	0.516	0.000	-3.000	2.000
JOYREAD	77,834	0.208	1.081	0.167	-2.732	2.657
SCREADCOMP	76,187	0.021	0.974	0.122	-2.440	1.884
SCREADDIFF	76,299	0.061	0.986	-0.070	-1.888	2.775
UNDREM	74,706	-0.038	1.008	0.100	-1.640	1.500
METASUM	74,864	-0.090	1.009	0.210	-1.720	1.360
METASPAM	74,408	0.009	1.006	-0.040	-1.410	1.330
EFFORT1	71,813	7.898	1.841	8.000	1.000	10.000

The average reading score was 525.63 ($SD = 99.97$), with a median of 530.78. ESCS and most reading indices were centred near zero, while average reported test effort was 7.90 on the 1-10 scale. Among students with valid repetition data, 6.8% had repeated a grade. Figure 1 shows that reading scores were approximately bell-shaped and that ESCS was concentrated around zero with a small lower tail.

Figure 1

Distributions of reading performance and ESCS. Vertical lines show the sample means.

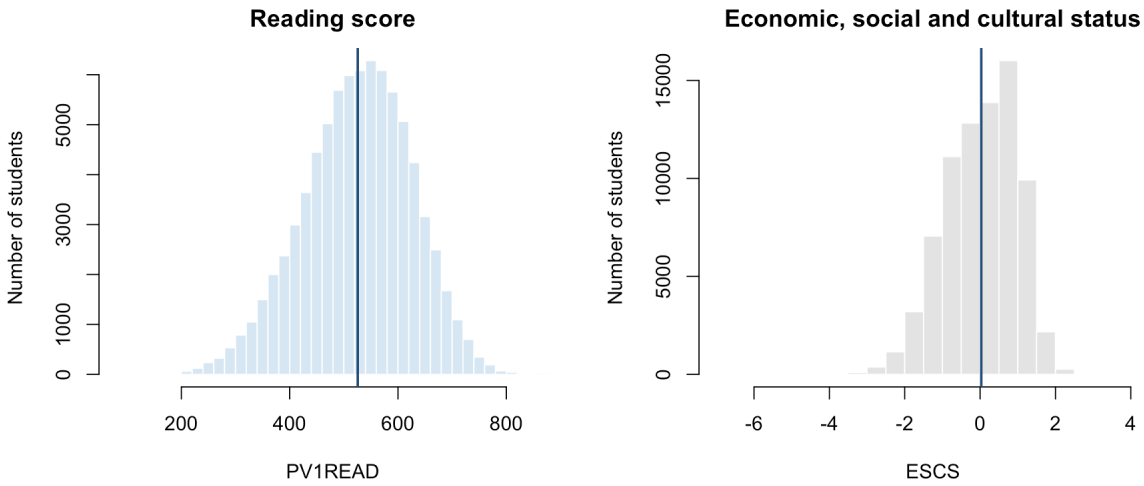


Table 3

Distribution of students across home-book categories

Books at home	N	**Percent**
0-10 books	9,284	11.9%
11-25 books	12,430	15.9%
26-100 books	25,609	32.7%
101-200 books	14,462	18.5%
201-500 books	11,300	14.4%
More than 500 books	5,170	6.6%

The largest group reported 26-100 books at home (32.7%). About 21.0% reported more than 200 books, while 27.7% reported 25 books or fewer.

Comparative Analysis

Gender comparison

Table 4

Reading performance by gender

Gender	N	M	SD
Female	39,424	539.52	93.96
Male	40,590	512.15	103.71

Levene's test was significant, $F(1, 80{,}012) = 410.26$, $p < .001$, so the equal-variance assumption was not met. Welch's t test found a statistically significant difference,

$t(79{,}629) = 39.15$, $p < .001$, 95% CI for the female-minus-male difference [26.01, 28.75]. Female students scored 27.38 points higher on average, so the gender null hypothesis was rejected.

Home-book comparison

Table 5
Reading performance by home-book category

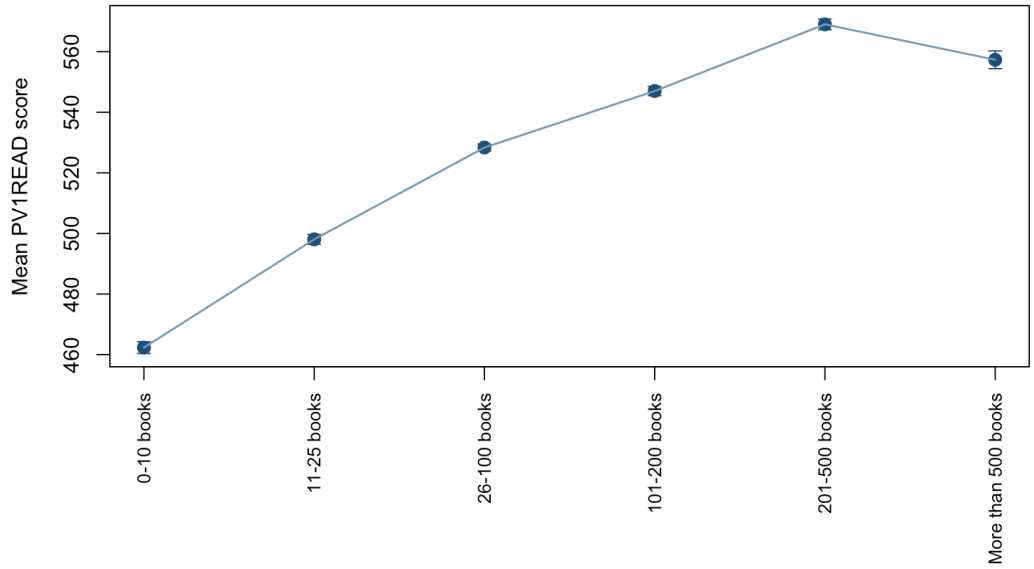
Books at home	N	M	SD
0-10 books	9,284	462.32	94.57
11-25 books	12,430	498.04	91.66
26-100 books	25,609	528.35	91.97
101-200 books	14,462	547.01	94.21
201-500 books	11,300	569.02	96.06
More than 500 books	5,170	557.32	107.17

Levene's test was significant, $F(5, 78{,}249) = 34.41$, $p < .001$. Welch's one-way ANOVA therefore replaced the equal-variance ANOVA. Reading means differed significantly, $F(5, 26{,}356) = 1761.80$, $p < .001$, and every Holm-adjusted pairwise comparison was significant (all adjusted $p < .001$).

Mean reading performance increased from 462.32 in the 0-10 books group to 569.02 in the 201-500 books group, then decreased to 557.32 in the more-than-500 group. The raw difference between the 0-10 and 201-500 groups was 106.70 points.

Figure 2

Mean reading scores by books at home. Error bars represent 95% confidence intervals.

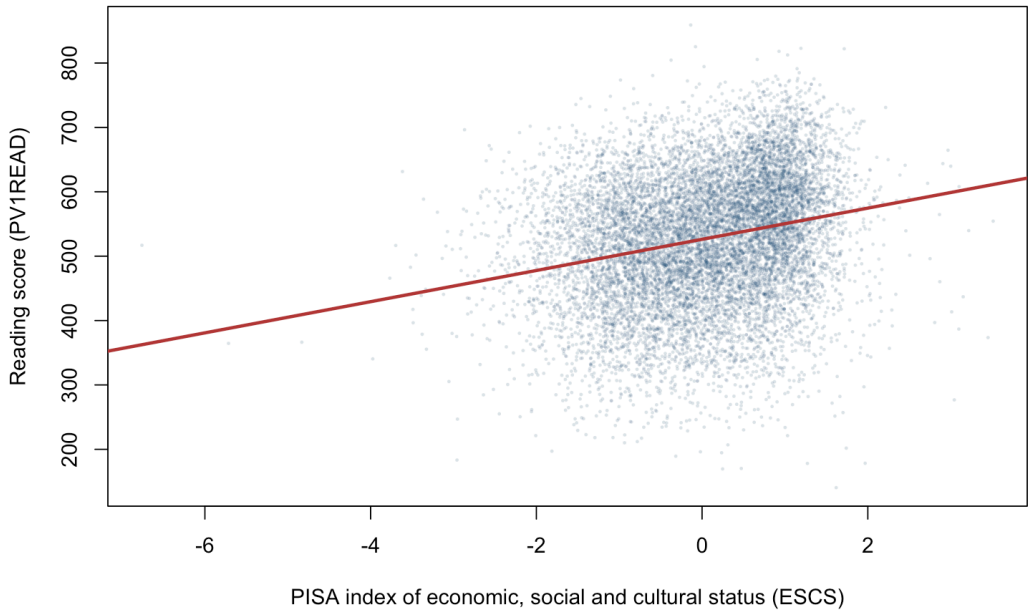


Correlation Analysis

ESCS had a positive and statistically significant correlation with reading performance, $r(78{,}191) = .232$, $p < .001$, 95% CI [.225, .238]. The association was weak to moderate: ESCS alone accounted for about 5.4% of the variation in PV1READ. Reading scores still varied widely at every ESCS level.

Figure 3

Relationship between ESCS and reading performance. A random sample of 12,000 complete cases is shown for readability; the line is based on all complete cases.



Multiple Regression Analysis

Table 6

Comparison of multiple linear regression models

Model	N	R^2	**Adjusted** R^2	**Residual** SE
1. Background and education context	63,284	.188	.188	86.44
2. + engagement and self-concept	63,284	.284	.284	81.19
3. + metacognitive strategies	63,284	.431	.431	72.36
4. + test effort	63,284	.437	.437	71.99

The background and education-context model explained 18.8% of reading-score variation. Reading engagement and self-concept increased R^2 to 28.4%. Metacognitive strategy knowledge produced the largest increase, raising R^2 to 43.1%. Test effort raised it to 43.7%. Every block significantly improved model fit (all $p < .001$). The final model was significant, $F(25, 63{,}258) = 1964.95$, $p < .001$, although 56.3% of the variation remained unexplained.

Because the prespecified regression model was additive, no interaction coefficient is reported.

Figure 4

R^2 across the four reading-performance models. All models use the same complete-case sample.

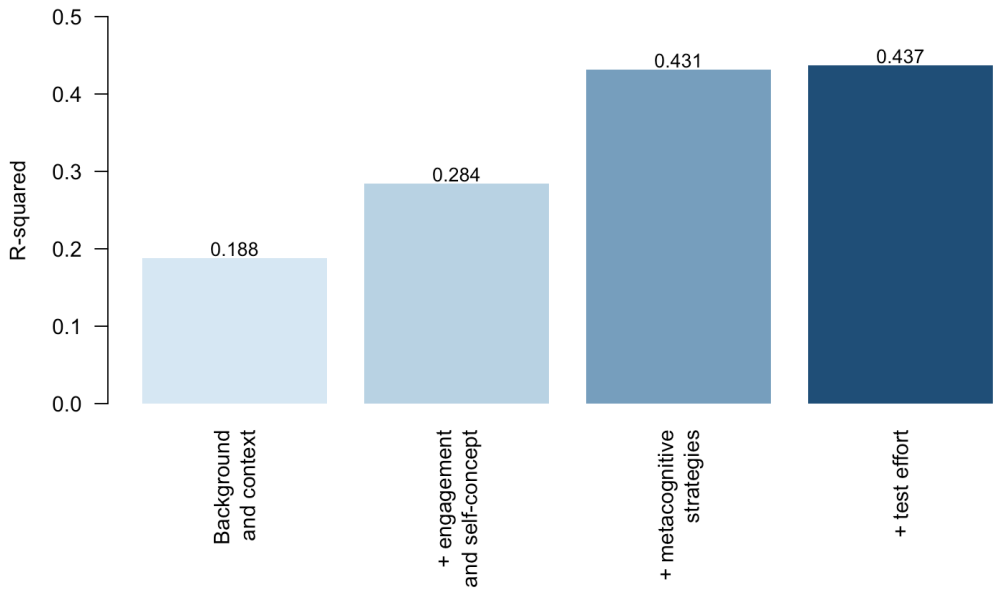


Table 7

Main coefficients from the full regression model

Predictor	b	**Robust** SE	**95% CI**	p
ESCS	10.52	0.39	[9.76, 11.29]	< .001
Books: 11-25	8.30	1.19	[5.96, 10.63]	< .001
Books: 26-100	19.37	1.10	[17.22, 21.52]	< .001
Books: 101-200	26.39	1.23	[23.97, 28.81]	< .001
Books: 201-500	34.39	1.35	[31.74, 37.03]	< .001
Books: >500	27.04	1.65	[23.80, 30.29]	< .001
Male	-0.73	0.60	[-1.91, 0.45]	.224
Relative grade placement	14.72	0.69	[13.37, 16.07]	< .001
Repeated a grade	-29.77	1.46	[-32.63, -26.92]	< .001
Reading enjoyment	8.90	0.35	[8.22, 9.58]	< .001
Perceived reading competence	12.62	0.41	[11.80, 13.43]	< .001
Perceived reading difficulty	-8.21	0.36	[-8.92, -7.50]	< .001
Understanding/remembering	8.65	0.34	[7.99, 9.32]	< .001
Summarising strategies	15.97	0.35	[15.28, 16.66]	< .001
Credibility-assessment strategies	25.25	0.32	[24.61, 25.88]	< .001
Test effort	4.29	0.19	[3.91, 4.66]	< .001

Note. Reference categories are 0-10 books, female and no grade repetition. Education-system

indicators were included but are not displayed. HC3 robust standard errors and confidence intervals are reported.

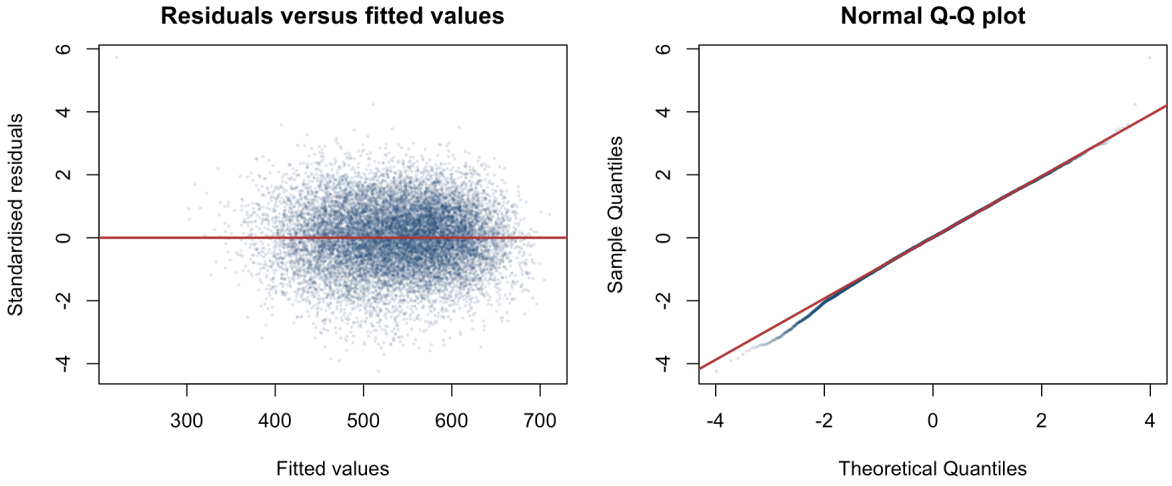
After adjustment, a one-unit increase in ESCS was associated with a 10.52-point increase in reading performance. Students in the 201-500 books category were predicted to score 34.39 points higher than those in the 0-10 books category. Relative grade placement was positively associated with reading, while grade repetition was associated with a 29.77-point lower score.

All seven reading-related predictors were significant in the expected directions. Reading enjoyment, perceived competence, knowledge of effective strategies and test effort were positively associated with performance; perceived reading difficulty was negatively associated. The raw female advantage did not remain after adjustment: the male coefficient was -0.73 and was not significant, $p = .224$. These coefficients describe conditional associations, not causal effects.

Adjusted generalised variance inflation factors ranged from 1.04 to 1.31, so multicollinearity was not a serious concern. The non-constant variance test was significant, $p < .001$, so HC3 robust standard errors were used. The maximum Cook's distance was .00214, far below 1, indicating that no single case dominated the model. Residual plots showed some heteroskedasticity and tail departures, consistent with the use of robust inference.

Figure 5

Diagnostic plots for the final regression model. A random sample of 15,000 residuals is shown for readability.



5. Discussion

Findings Interpretation

The bivariate analyses showed a 27.38-point female advantage, clear differences across home-book categories and a positive ESCS-reading correlation. These findings describe substantial social patterning, but the multiple regression gives a more complete account. For

RQ4a, the final model explained 43.7% of the variation in reading performance, with metacognitive strategy knowledge producing the largest increase in R^2. For RQ4b, every reading-related predictor remained significant after background and education context were controlled.

The model comparison is more informative than a single R^2 value. Background and education context explained 18.8% of variation. Engagement and self-concept added 9.6 percentage points, while metacognitive strategy knowledge added 14.7 percentage points. Test effort added a smaller 0.6 percentage points. The final R^2 is considerably stronger than the background-only model, but it still leaves 56.3% of individual differences unexplained.

Home books and ESCS remained significant, although their adjusted associations were smaller than the raw group differences. The 106.70-point raw difference between the 0-10 and 201-500 book groups decreased to 34.39 points in the full model. This suggests that part of the raw books pattern overlaps with education system, grade progression, reading engagement, self-concept and strategy knowledge.

The raw gender difference also changed after adjustment. Female students scored substantially higher in the bivariate comparison, but gender was no longer significant in the final model. This does not mean that gender is unimportant. It means that, within this sample, the observed gender gap overlapped with differences in the included reading-related and contextual variables.

Knowledge of effective reading strategies was especially important. Higher scores for summarising and credibility assessment were associated with higher reading performance, as were enjoyment and perceived competence. Perceived difficulty was negatively associated. These patterns fit the PISA reading framework, but the cross-sectional data cannot show that changing a questionnaire index would cause a particular score increase.

Limitations

Several limitations affect the interpretation. First, the trimmed dataset contains ten relatively high-performing education systems rather than all PISA 2018 participants. Second, the file does not include student sampling weights or replicate weights. OECD guidance states that official PISA estimates should use these weights to account for the sampling design (OECD, n.d.). The present unweighted results should therefore be interpreted as sample-based course analyses.

Third, only PV1READ is used, whereas official PISA analysis normally combines all plausible values. Fourth, school identifiers are unavailable, so students are not modelled as nested within schools. Fifth, most predictors are self-reported and may contain response bias. Test effort was reported after the assessment and may partly reflect how students believed they had performed.

Finally, the cross-sectional design cannot establish causal order. Strong readers may

develop greater confidence, enjoyment and strategy knowledge, rather than these characteristics only producing higher scores. The model leaves 56.3% of variation unexplained. Prior reading achievement, cognitive skills, language background, teacher quality, school resources, peer influences and family reading practices may account for additional variation.

Implications for Social Research

The results suggest that reading inequality is connected to both family resources and how students engage with texts. Schools cannot directly change family socioeconomic status, but they can provide access to varied reading materials and explicitly teach strategies for summarising, monitoring understanding and judging the credibility of information. Activities that build reading confidence and enjoyment may also be relevant. These are practical possibilities suggested by the associations, not interventions proven by this analysis.

6. Conclusion

Key Results Recap

This project examined how family background, education context and reading-related characteristics were associated with reading performance in a trimmed PISA 2018 dataset. Reading scores differed across gender and home-book groups, and ESCS was positively correlated with reading performance.

The final multiple linear regression explained 43.7% of reading-score variation, compared with 18.8% for the background and education-context model. Metacognitive strategy knowledge produced the largest improvement in explanatory power. ESCS, home books, grade progression, reading enjoyment, self-concept, strategy knowledge and test effort remained associated with reading performance, while the adjusted gender coefficient was not significant.

Further Research

Future research should use all plausible values, appropriate sampling weights and multilevel models, and should examine longitudinal evidence before making causal claims.

7. References

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OpenAI. (2026). *ChatGPT (GPT-5 version)* [Large language model]. https://chatgpt.com/

Use of Generative Artificial Intelligence

OpenAI's ChatGPT was used to help construct the R code framework, restructure the multiple linear regression analysis, troubleshoot code, and polish the language of the report. AI was not used as a source of data or empirical evidence. The variables, code, statistical results, citations, and interpretations were checked against the supplied dataset and the final R output by the authors (OpenAI, 2026).

Appendix A: R Code

The complete R script is also submitted as a separate file. The code below covers data preparation, descriptive statistics, comparative tests, correlation, regression and diagnostics.

R code

```
# UFUG1811 Final Project – R Code Submission
# Family background, reading processes, and reading performance
#
# Reproducibility instructions:
# 1. Put this script and PISA2018_top10_new.sav in the same folder.
# 2. Set the R working directory to that folder.
# 3. Run the complete script from the first line.
#
# Required packages: haven and car.
# Results are written to the project_output folder.

output_dir <- "project_output"

library(haven)
library(car)

dir.create(output_dir, recursive = TRUE, showWarnings = FALSE)

# Read only the variables used in this project.
pisa <- read_sav(
  "PISA2018_top10_new.sav",
  col_select = c(
    CNT, ST004D01T, ST013Q01TA, ESCS, PV1READ,
    GRADE, REPEAT, JOYREAD, SCREADCOMP, SCREADDIFF,
    UNDREM, METASUM, METASPAM, EFFORT1
  )
)

# Recode labelled categorical variables.
pisa$gender <- factor(
  pisa$ST004D01T,
  levels = c(1, 2),
  labels = c("Female", "Male")
)

pisa$books <- factor(
  pisa$ST013Q01TA,
  levels = 1:6,
  labels = c(
    "0–10 books", "11–25 books", "26–100 books",
    "101–200 books", "201–500 books", "More than 500 books"
  )
)

pisa$country <- factor(pisa$CNT)

pisa$repeat_grade <- factor(
  pisa$REPEAT,
  levels = c(0, 1),
  labels = c("No", "Yes")
```

```
)

# ------------------------------------------------------------------------------
# 1. Descriptive analysis
# ------------------------------------------------------------------------------
continuous_variables <- c(
  "PV1READ", "ESCS", "GRADE", "JOYREAD", "SCREADCOMP",
  "SCREADDIFF", "UNDREM", "METASUM", "METASPAM", "EFFORT1"
)

continuous_stats <- data.frame(
  Variable = continuous_variables,
  N = sapply(pisa[continuous_variables], function(x) sum(!is.na(x))),
  Missing = sapply(pisa[continuous_variables], function(x) sum(is.na(x))),
  Mean = sapply(pisa[continuous_variables], mean, na.rm = TRUE),
  SD = sapply(pisa[continuous_variables], sd, na.rm = TRUE),
  Min = sapply(pisa[continuous_variables], min, na.rm = TRUE),
  Median = sapply(pisa[continuous_variables], median, na.rm = TRUE),
  Max = sapply(pisa[continuous_variables], max, na.rm = TRUE)
)
rownames(continuous_stats) <- NULL

missing_stats <- data.frame(
  Variable = c(
    "PV1READ", "ESCS", "Books", "Gender", "Country", "GRADE", "REPEAT",
    "JOYREAD", "SCREADCOMP", "SCREADDIFF", "UNDREM", "METASUM",
    "METASPAM", "EFFORT1"
  ),
  N_nonmissing = c(
    sum(!is.na(pisa$PV1READ)),
    sum(!is.na(pisa$ESCS)),
    sum(!is.na(pisa$books)),
    sum(!is.na(pisa$gender)),
    sum(!is.na(pisa$country)),
    sum(!is.na(pisa$GRADE)),
    sum(!is.na(pisa$repeat_grade)),
    sum(!is.na(pisa$JOYREAD)),
    sum(!is.na(pisa$SCREADCOMP)),
    sum(!is.na(pisa$SCREADDIFF)),
    sum(!is.na(pisa$UNDREM)),
    sum(!is.na(pisa$METASUM)),
    sum(!is.na(pisa$METASPAM)),
    sum(!is.na(pisa$EFFORT1))
  )
)
missing_stats$Missing <- nrow(pisa) - missing_stats$N_nonmissing
missing_stats$Missing_percent <- 100 * missing_stats$Missing / nrow(pisa)

book_freq <- as.data.frame(table(pisa$books, useNA = "no"))
names(book_freq) <- c("Books", "N")
book_freq$Percent <- 100 * book_freq$N / sum(book_freq$N)

repeat_freq <- as.data.frame(table(pisa$repeat_grade, useNA = "no"))
names(repeat_freq) <- c("Repeated_grade", "N")
repeat_freq$Percent <- 100 * repeat_freq$N / sum(repeat_freq$N)
```

```
gender_stats <- aggregate(
  PV1READ ~ gender,
  data = subset(pisa, !is.na(PV1READ) & !is.na(gender)),
  FUN = function(x) c(N = length(x), Mean = mean(x), SD = sd(x))
)
gender_stats <- data.frame(
  Gender = gender_stats$gender,
  N = gender_stats$PV1READ[, "N"],
  Mean = gender_stats$PV1READ[, "Mean"],
  SD = gender_stats$PV1READ[, "SD"]
)

book_stats <- aggregate(
  PV1READ ~ books,
  data = subset(pisa, !is.na(PV1READ) & !is.na(books)),
  FUN = function(x) {
    c(N = length(x), Mean = mean(x), SD = sd(x), SE = sd(x) / sqrt(length(x)))
  }
)
book_stats <- data.frame(
  Books = book_stats$books,
  N = book_stats$PV1READ[, "N"],
  Mean = book_stats$PV1READ[, "Mean"],
  SD = book_stats$PV1READ[, "SD"],
  SE = book_stats$PV1READ[, "SE"]
)
book_stats$Lower95 <- book_stats$Mean - 1.96 * book_stats$SE
book_stats$Upper95 <- book_stats$Mean + 1.96 * book_stats$SE

# ------------------------------------------------------------------------------
# 2. Comparative analyses
# ------------------------------------------------------------------------------
t_data <- subset(pisa, !is.na(PV1READ) & !is.na(gender))
gender_levene <- leveneTest(PV1READ ~ gender, data = t_data, center = median)
gender_t <- t.test(PV1READ ~ gender, data = t_data, var.equal = FALSE)

anova_data <- subset(pisa, !is.na(PV1READ) & !is.na(books))
books_levene <- leveneTest(PV1READ ~ books, data = anova_data, center = median)
books_welch <- oneway.test(PV1READ ~ books, data = anova_data, var.equal = FALSE)
books_pairwise <- pairwise.t.test(
  anova_data$PV1READ,
  anova_data$books,
  p.adjust.method = "holm",
  pool.sd = FALSE
)

# ------------------------------------------------------------------------------
# 3. Correlation analysis
# ------------------------------------------------------------------------------
cor_data <- subset(pisa, !is.na(PV1READ) & !is.na(ESCS))
escs_cor <- cor.test(cor_data$ESCS, cor_data$PV1READ, method = "pearson")

# ------------------------------------------------------------------------------
# 4. Multiple linear regression
# ------------------------------------------------------------------------------
# All four models use the same complete-case sample so that R-squared values
```

```
# can be compared fairly.
reg_data <- subset(
  pisa,
  complete.cases(
    PV1READ, ESCS, books, gender, country, GRADE, repeat_grade,
    JOYREAD, SCREADCOMP, SCREADDIFF,
    UNDREM, METASUM, METASPAM, EFFORT1
  )
)

model_1 <- lm(
  PV1READ ~ ESCS + books + gender + country + GRADE + repeat_grade,
  data = reg_data
)
model_2 <- lm(
  PV1READ ~ ESCS + books + gender + country + GRADE + repeat_grade +
    JOYREAD + SCREADCOMP + SCREADDIFF,
  data = reg_data
)
model_3 <- lm(
  PV1READ ~ ESCS + books + gender + country + GRADE + repeat_grade +
    JOYREAD + SCREADCOMP + SCREADDIFF +
    UNDREM + METASUM + METASPAM,
  data = reg_data
)
model_4 <- lm(
  PV1READ ~ ESCS + books + gender + country + GRADE + repeat_grade +
    JOYREAD + SCREADCOMP + SCREADDIFF +
    UNDREM + METASUM + METASPAM + EFFORT1,
  data = reg_data
)

model_summary <- data.frame(
  Model = c(
    "Background and education context",
    "+ reading engagement and self-concept",
    "+ metacognitive strategy knowledge",
    "+ test effort"
  ),
  N = sapply(list(model_1, model_2, model_3, model_4), nobs),
  R2 = sapply(
    list(model_1, model_2, model_3, model_4),
    function(m) summary(m)$r.squared
  ),
  Adjusted_R2 = sapply(
    list(model_1, model_2, model_3, model_4),
    function(m) summary(m)$adj.r.squared
  ),
  Residual_SE = sapply(
    list(model_1, model_2, model_3, model_4),
    sigma
  )
)

block_comparison <- anova(model_1, model_2, model_3, model_4)
```

```
# The residual-variance test is significant in this large sample. Coefficients
# are unchanged, but HC3 heteroskedasticity-robust standard errors are used for
# the final coefficient tests and confidence intervals.
robust_vcov <- hccm(model_4, type = "hc3")
robust_se <- sqrt(diag(robust_vcov))
robust_t <- coef(model_4) / robust_se
robust_p <- 2 * pt(abs(robust_t), df.residual(model_4), lower.tail = FALSE)
critical_t <- qt(0.975, df.residual(model_4))

robust_coef_table <- data.frame(
  Term = names(coef(model_4)),
  Estimate = unname(coef(model_4)),
  Robust_SE = unname(robust_se),
  t_value = unname(robust_t),
  p_value = unname(robust_p),
  Lower95 = unname(coef(model_4) - critical_t * robust_se),
  Upper95 = unname(coef(model_4) + critical_t * robust_se)
)

# Regression diagnostics.
vif_raw <- vif(model_4)
if (is.matrix(vif_raw)) {
  vif_table <- data.frame(
    Term = rownames(vif_raw),
    GVIF = vif_raw[, "GVIF"],
    Df = vif_raw[, "Df"],
    Adjusted_GVIF = vif_raw[, "GVIF^(1/(2*Df))"]
  )
} else {
  vif_table <- data.frame(Term = names(vif_raw), VIF = as.numeric(vif_raw))
}

ncv_result <- ncvTest(model_4)
cook <- cooks.distance(model_4)
diagnostic_stats <- data.frame(
  Metric = c(
    "Maximum Cook's distance",
    "Cases with Cook's distance > 4/n",
    "Non-constant variance test chi-square",
    "Non-constant variance test p-value"
  ),
  Value = c(
    max(cook),
    sum(cook > 4 / nobs(model_4)),
    unname(ncv_result$ChiSquare),
    ncv_result$p
  )
)

# ------------------------------------------------------------------------------
```

Appendix B: Selected R Output

The following console output records the principal inferential tests and the full regression model.

Selected R output

```
GENDER LEVENE TEST
Levene's Test for Homogeneity of Variance (center = median)
         Df F value    Pr(>F)
group     1  410.26 < 2.2e-16 ***
      80012
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

WELCH T-TEST

        Welch Two Sample t-test

data:  PV1READ by gender
t = 39.153, df = 79629, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
 26.00763 28.74870
sample estimates:
mean in group Female   mean in group Male
            539.5244             512.1462


BOOKS LEVENE TEST
Levene's Test for Homogeneity of Variance (center = median)
         Df F value    Pr(>F)
group     5  34.412 < 2.2e-16 ***
      78249
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

WELCH ONE-WAY ANOVA

        One-way analysis of means (not assuming equal variances)

data:  PV1READ and books
F = 1761.8, num df = 5, denom df = 26356, p-value < 2.2e-16


PAIRWISE WELCH TESTS WITH HOLM ADJUSTMENT

        Pairwise comparisons using t tests with non-pooled SD

data:  anova_data$PV1READ and anova_data$books

                   0-10 books 11-25 books 26-100 books 101-200 books
11-25 books        < 2e-16    -           -            -
26-100 books       < 2e-16    < 2e-16     -            -
101-200 books      < 2e-16    < 2e-16     < 2e-16      -
201-500 books      < 2e-16    < 2e-16     < 2e-16      < 2e-16
More than 500 books < 2e-16    < 2e-16     < 2e-16      9.8e-10
```

```
                   201–500 books
11–25 books        –
26–100 books       –
101–200 books      –
201–500 books      –
More than 500 books 4.0e-11

P value adjustment method: holm

PEARSON CORRELATION

        Pearson's product-moment correlation

data:  cor_data$ESCS and cor_data$PV1READ
t = 66.62, df = 78191, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.2251175 0.2383829
sample estimates:
     cor
0.231761


REGRESSION MODEL COMPARISON
                                  Model     N        R2 Adjusted_R2 Residual_SE
1      Background and education context 63284 0.1884006   0.1881697    86.44423
2 + reading engagement and self-concept 63284 0.2840964   0.2838587    81.19004
3    + metacognitive strategy knowledge 63284 0.4313885   0.4311728    72.35917
4                          + test effort 63284 0.4371148   0.4368923    71.99447

NESTED MODEL TESTS
Analysis of Variance Table

Model 1: PV1READ ~ ESCS + books + gender + country + GRADE + repeat_grade
Model 2: PV1READ ~ ESCS + books + gender + country + GRADE + repeat_grade +
    JOYREAD + SCREADCOMP + SCREADDIFF
Model 3: PV1READ ~ ESCS + books + gender + country + GRADE + repeat_grade +
    JOYREAD + SCREADCOMP + SCREADDIFF + UNDREM + METASUM + METASPAM
Model 4: PV1READ ~ ESCS + books + gender + country + GRADE + repeat_grade +
    JOYREAD + SCREADCOMP + SCREADDIFF + UNDREM + METASUM + METASPAM +
    EFFORT1
  Res.Df       RSS Df Sum of Sq       F    Pr(>F)
1  63265 472754375
2  63262 417011857  3  55742518 3584.82 < 2.2e-16 ***
3  63259 331214615  3  85797241 5517.65 < 2.2e-16 ***
4  63258 327879081  1   3335535  643.53 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

FINAL REGRESSION MODEL

Call:
lm(formula = PV1READ ~ ESCS + books + gender + country + GRADE +
    repeat_grade + JOYREAD + SCREADCOMP + SCREADDIFF + UNDREM +
    METASUM + METASPAM + EFFORT1, data = reg_data)
```

```
Residuals:
    Min      1Q  Median      3Q     Max
-382.54  -46.60    1.25   48.48  411.69

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)             469.4396     1.6747 280.307  < 2e-16 ***
ESCS                     10.5247     0.3745  28.103  < 2e-16 ***
books11-25 books          8.2955     1.1529   7.195 6.31e-13 ***
books26-100 books        19.3722     1.0619  18.243  < 2e-16 ***
books101-200 books       26.3905     1.2002  21.989  < 2e-16 ***
books201-500 books       34.3856     1.3018  26.415  < 2e-16 ***
booksMore than 500 books 27.0443     1.5714  17.211  < 2e-16 ***
genderMale               -0.7306     0.6013  -1.215   0.2244
countryEST                7.6682     1.2576   6.097 1.08e-09 ***
countryFIN                1.6397     1.2428   1.319   0.1870
countryHKG               47.0599     1.2633  37.252  < 2e-16 ***
countryIRL              -10.3253     1.2865  -8.026 1.02e-15 ***
countryKOR                7.1525     1.1607   6.162 7.21e-10 ***
countryMAC               43.1059     1.4439  29.854  < 2e-16 ***
countryPOL                2.4469     1.2606   1.941   0.0523 .
countryQCI               36.6123     1.0520  34.802  < 2e-16 ***
countrySGP               34.0682     1.1660  29.218  < 2e-16 ***
GRADE                    14.7164     0.6820  21.579  < 2e-16 ***
repeat_gradeYes         -29.7745     1.3460 -22.120  < 2e-16 ***
JOYREAD                   8.8986     0.3459  25.725  < 2e-16 ***
SCREADCOMP               12.6169     0.3811  33.106  < 2e-16 ***
SCREADDIFF               -8.2075     0.3311 -24.786  < 2e-16 ***
UNDREM                    8.6546     0.3344  25.880  < 2e-16 ***
METASUM                  15.9672     0.3492  45.728  < 2e-16 ***
METASPAM                 25.2475     0.3220  78.418  < 2e-16 ***
EFFORT1                   4.2872     0.1690  25.368  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 71.99 on 63258 degrees of freedom
Multiple R-squared:  0.4371,        Adjusted R-squared:  0.4369
F-statistic:  1965 on 25 and 63258 DF,  p-value: < 2.2e-16


HC3 ROBUST COEFFICIENTS
                       Term    Estimate Robust_SE    t_value       p_value
1               (Intercept) 469.4395786 1.8467192 254.201927  0.000000e+00
2                      ESCS  10.5246804 0.3902324  26.970288 2.642689e-159
3          books11-25 books   8.2954839 1.1894225   6.974379  3.102272e-12
4         books26-100 books  19.3721765 1.0981988  17.639955  1.784408e-69
5        books101-200 books  26.3904643 1.2330734  21.402184 2.920365e-101
6        books201-500 books  34.3855913 1.3496463  25.477483 1.840121e-142
7  booksMore than 500 books  27.0443220 1.6536395  16.354424  5.373890e-60
8                genderMale  -0.7306156 0.6002128  -1.217261  2.235095e-01
9                countryEST   7.6681567 1.2200787   6.284969  3.300494e-10
10               countryFIN   1.6397078 1.1688634   1.402822  1.606747e-01
11               countryHKG  47.0598969 1.3272065  35.457856 1.055229e-272
12               countryIRL -10.3252753 1.1971305  -8.625021  6.553516e-18
13               countryKOR   7.1524549 1.2106889   5.907756  3.485663e-09
14               countryMAC  43.1058513 1.5129086  28.492039 1.957268e-177
```

```
15               countryPOL   2.4468674 1.2548480   1.949931  5.118872e-02
16               countryQCI  36.6122519 1.0466056  34.981899 1.476450e-265
17               countrySGP  34.0681959 1.2712050  26.799923 2.463044e-157
18                    GRADE  14.7163614 0.6892828  21.350251 8.813436e-101
19          repeat_gradeYes -29.7744749 1.4551626 -20.461269  9.534429e-93
20                  JOYREAD   8.8985700 0.3487157  25.518124 6.587156e-143
21               SCREADCOMP  12.6169187 0.4143717  30.448313 3.671903e-202
22               SCREADDIFF  -8.2074519 0.3623907 -22.648072 4.116337e-113
23                   UNDREM   8.6545759 0.3379762  25.607054 6.918831e-144
24                  METASUM  15.9671879 0.3529737  45.236199  0.000000e+00
25                 METASPAM  25.2474526 0.3226884  78.240955  0.000000e+00
26                  EFFORT1   4.2872153 0.1919187  22.338702 4.155218e-110
        Lower95     Upper95
1  465.82000621 473.0591510
2    9.75982429  11.2895365
3    5.96421396  10.6267539
4   17.21970525  21.5246477
5   23.97363859  28.8072900
6   31.74028252  37.0309002
7   23.80318609  30.2854578
8   -1.90703347   0.4458023
9    5.27680049  10.0595128
10  -0.65126613   3.9306817
11  44.45857021  49.6612236
12 -12.67165281  -7.9788977
13   4.77950292   9.5254069
14  40.14054814  46.0711545
15  -0.01263652   4.9063713
16  34.56090337  38.6636005
17  31.57663225  36.5597595
18  13.36536604  16.0673567
19 -32.62659585 -26.9223540
20   8.21508672   9.5820532
21  11.80474962  13.4290879
22  -8.91773835  -7.4971655
23   7.99214198   9.3170099
24  15.27535899  16.6590169
25  24.61498272  25.8799224
26   3.91105431   4.6633763

MULTICOLLINEARITY
                     Term     GVIF Df Adjusted_GVIF
ESCS                 ESCS 1.544910  1      1.242944
books               books 1.537956  5      1.043985
gender             gender 1.103641  1      1.050543
country           country 2.123197  9      1.042716
GRADE               GRADE 1.448891  1      1.203699
repeat_grade repeat_grade 1.318701  1      1.148347
JOYREAD           JOYREAD 1.712002  1      1.308435
SCREADCOMP     SCREADCOMP 1.661882  1      1.289140
SCREADDIFF     SCREADDIFF 1.281667  1      1.132107
UNDREM             UNDREM 1.374807  1      1.172522
METASUM           METASUM 1.492580  1      1.221712
METASPAM         METASPAM 1.271139  1      1.127448
EFFORT1           EFFORT1 1.110412  1      1.053761
```

```
NON-CONSTANT VARIANCE TEST
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 307.2661, Df = 1, p = < 2.22e-16

INFLUENCE DIAGNOSTICS
                                Metric        Value
1              Maximum Cook's distance 2.138428e-03
2      Cases with Cook's distance > 4/n 3.338000e+03
3 Non-constant variance test chi-square 3.072661e+02
4    Non-constant variance test p-value 8.606033e-69
```